1 OCR High School uses a computer system to store data about students' conduct. The system records good conduct as a positive number and poor conduct as a negative number. A TRUE or FALSE value is also used to record whether or not a letter has been sent home about each incident.

An example of the data held in this system is shown below in Fig. 1:

StudentName	Detail	Points	LetterSent
Kirstie	Homework forgotten	–2	FALSE
Byron	Good effort in class	1	TRUE
Grahame	100% in a test	2	FALSE
Marian	Bullying	–3	TRUE

Fig. 1

A single record from this database table is read into a program that uses an array with the identifier `studentdata`. An example of this array is shown below:

```
studentdata = ["Kirstie", "Homework forgotten", "–2", "FALSE"]
```

The array is zero based, so `studentdata[0]` holds the value "Kirstie".

Write an algorithm that will identify whether the data in the `studentdata` array shows that a letter has been sent home or not for the student. The algorithm should then output either "sent" (if a letter has been sent) or "not sent" (if a letter has not been sent).

[4]

2 OCR Drones flies goods around the country using drones.

A pilot is allowed to fly a maximum of 9 hours per day. The pilot code and the hours flown for each flight in one day are stored in a 2D array of strings with the identifier `journeys`.

The data in `journeys` is shown.

	[0]	[1]
[0]	"SM720"	"4.5"
[1]	"GC730"	"3"
[2]	"JP670"	"2"
[3]	"GC730"	"3.5"
[4]	"CC200"	"9"
[5]	"RY320"	"12"

The value of `journeys[3,1]` is `"3.5"`.

Create a function, `pilotValid()`, that:

- takes a pilot code as a parameter
- uses the 2D array `journeys` to calculate the number of hours that the pilot has flown
- returns `valid` if the pilot has flown 9 hours or fewer or `warning` if the pilot has flown for more than 9 hours
- uses casting where needed.

You must use either:
- OCR Exam Reference Language, or
- A high-level programming language that you have studied.

[6]

3 OCR Security Services need to identify the total number of seconds the sensors have been activated on a specific date.

The data from the database table `events` is imported into the program written in a highlevel programming language.

The program stores the data in a two-dimensional (2D) string array with the identifier `arrayEvents`

The data to be stored is shown in the table.

Date	**SensorID**	**SensorType**	**Length**
05/02/2023	WS2	Window	38
05/02/2023	MS1	Motion	2
06/02/2023	DS3	Door	1
06/02/2023	MS2	Motion	3
06/02/2023	MS1	Motion	2
07/02/2023	WS1	Window	24
07/02/2023	DS1	Door	1

In this table, the value of `events[1, 1]` contains "`MS1`".

(i) An array can only store data of one data type. Any non-string data must be converted to a string before storing in the array.

Identify the process that converts integer data to string data.

____ **[1]**

(ii) Write a program that:

- asks the user to input a date
- totals the number of seconds sensors have been activated on the date input
- outputs the calculated total in an appropriate message including the date, for example:
  `Sensors were activated for 40 seconds on 05/02/2023`

You must use **either**:
- OCR Exam Reference Language, **or**
- A high-level programming language that you have studied.

____

[6]

4 A hotel has nine rooms that are numbered from room 0 to room 8.

The number of people currently staying in each room is stored in an array with the identifier `room`.

The index of `room` represents the room number.

Array `room`

Index	0	1	2	3	4	5	6	7	8
Data	2	1	3	2	1	0	0	4	1

The following program counts how many people are currently staying in the hotel.

```
for count = 1 to 8

    total = 0

    total = total + room[count]

next count

print(total)
```

When tested, the program is found to contain **two** logic errors.

Describe how the program can be refined to remove these logic errors.

[2]

5 OCRBlocks is a game played on a 5 × 5 grid. Players take it in turns to place blocks on the board. The board is stored as a two-dimensional (2D) array with the identifier `gamegrid`

Fig. 6.1 shows that players A and B have placed three blocks each so far.

	0	1	2	3	4
0	A			B	
1					
2		B			
3	A		B		
4			A		

Fig. 6.1

The function `checkblock()` checks whether a square on the board has been filled. When `checkblock(4,2)` is called, the value "A" is returned.

```
function checkblock(r,c)
  if gamegrid[r,c] == "A" or gamegrid[r,c] == "B" then
    outcome = gamegrid[r,c]
  else
    outcome = "FREE"
  endif
  return outcome
endfunction
```

Give the returned value when the following statements are called.

Function call	Returned value
`checkblock(2,1)`	
`checkblock(3,0)`	
`checkblock(2,3)`	

[3]

6 Data for one week (Monday to Friday) is stored in a 2D array with the identifier `minsPlayed`.

The following table shows part of this array, containing 4 students.

			Students			
			Stuart	**Wes**	**Victoria**	**Dan**
			0	**1**	**2**	**3**
Days of the week	**Mon**	**0**	60	30	45	0
	Tue	**1**	180	60	0	60
	Wed	**2**	200	30	0	20
	Thu	**3**	60	10	15	15
	Fri	**4**	100	35	30	45

The teacher wants to output the number of minutes Dan (column index 3) played computer games on Wednesday (row index 2). The following code is written:

```
print(minsPlayed[3,2])
```

Write program code to output the number of minutes that Stuart played computer games on Friday.

You must use **either**:

- OCR Exam Reference Language, **or**
- a high-level programming language that you have studied.

---

[1]

7(a) Heath is researching how long, to the nearest minute, each student in his class spends playing computer games in one week (Monday to Friday). He is storing the data in a 2D array.

Fig. 2 shows part of the array, with 4 students.

Fig. 2

Students

Days of the week	0	1	2	3
0	60	30	45	0
1	180	60	0	60
2	200	30	0	20
3	60	10	15	15
4	100	35	30	45

For example, student 1, on Monday (day 0), played 30 minutes of computer games.

Heath wants to output the number of minutes student 3 played computer games on Wednesday (day 2). He writes the code:

```
print (hoursPlayed[3,2])
```

The output is 20.

(i) Write the code to output the number of minutes student 0 played computer games on Wednesday.

____

____ **[1]**

(ii) State the output if Heath runs the code:

```
print (hoursPlayed[2,1])
```

____ **[1]**

(iii) State the output if Heath runs the code:

```
print (hoursPlayed[3,1] + hoursPlayed[3,2])
```

____

[1]

(iv) Write an algorithm to output the total number of minutes student 0 played computer games from Monday (day 0) to Friday (day 4).

[3]

(b) Heath needs to work out the average number of minutes spent playing computer games each day for the class, which contains 30 students. Write an algorithm to output the average number of minutes the whole class spends playing computer games each day.

[8]

END OF QUESTION PAPER

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>1</td><td></td><td></td><td>1 mark per bullet, max 4

• Selection(IF) used
• Comparing studentdata[3]…
• …with “TRUE” or “FALSE” // TRUE or FALSE
• Correct outputs (“sent” and “not sent”)</td><td>4</td><td>Example algorithm

<code>if studentdata[3] == “TRUE” then</code>
<code> print “sent”</code>
<code>else</code>
<code> print “not sent”</code>
<code>end if</code>

Bullet point 3 can only be awarded If an attempt is made at identifying <code>studentdata</code> (e.g. with the wrong index or no index). Do not allow simply comparing anything with True / False.

Bullet point 3 can be implicit.

Capitalisation not important.

“Sent” and “not sent” do not have to be exactly this – can be alternative message conveying same idea.

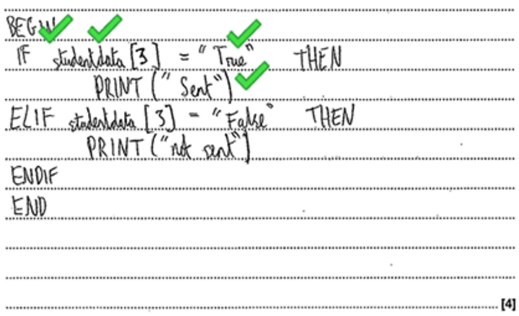
<u>Examiner’s Comments</u>

This question was generally well answered, at least in part, by candidates of all abilities. The use of selection (IF) is obviously well understood and this, together with the linked multiple outcomes of outputting ‘sent’ or ‘not sent’ were the most commonly credited mark points. However, despite the question providing very specific guidance on how to access the array, many candidates did not even attempt to do this.

Where the array was accessed correctly, a common mistake was to access the wrong element, usually index 4 rather than index 3. Arrays in questions on this specification will always be zero based and this fact was</td></tr>
</table>

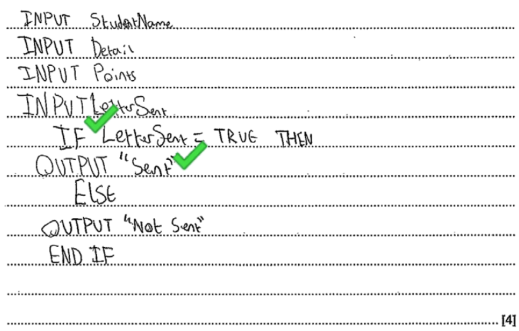
<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td>
also given in the question by means of an example. Candidates should be encouraged to read all parts of a question before beginning to attempt it.

Exemplar 1



In the above exemplar response, the candidate has achieved full marks by accessing the correct element in the array and using this to decide if the letter should be sent.

Exemplar 2



In the above exemplar response, the candidate has not attempted to access the array at all, instead asking the user to input a value; this does not do what is required and so is only given 2 of the 4 marks (for a valid use of selection and outputting a message correctly).
</td></tr>
</table>

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>4</td><td></td></tr>
<tr><td>2</td><td></td><td></td><td>One mark per bullet point to max 6

• Function header <code>pilotValid()</code> with (at least one) parameter
• Checks array element [i,0] for each item…
• …calculates total
• …casting to float / real
• Checks if 9 hours or fewer
• Returns a value
• …returns "warning" and "valid" correctly</td><td>6
(AO3 2b)</td><td>Example answer :
<pre>function
pilotValid(pilotCode)
  total = 0
  status = ""
  for i = 0 to 5
      if journeys[i,0] ==
pilotCode then
         temp =
float(journeys[i,1])
         total = total + temp
      endif
  next i
  if total > 9 then
      status = "warning"
  else
      status = "valid"
  endif
  return status
endfunction</pre>
Do not allow casting to integer for BP4, data shows some journeys have decimal places.

Allow FT for BP5 if attempt made at calculation.
Allow FT for BP7 if value is output instead of returned.</td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>6</td><td></td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
3		i	1 mark for: • Casting / cast	1 (AO3 2a)	Accept type casting Do not accept conversion. Do not accept examples of casting. **Examiner's Comments** The use of the term "casting" to convert one data type to another is now well known and understood by candidates. This is given and referred to in the J277 specification and is essential knowledge.

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td></td><td></td><td>ii</td><td>1 mark each to max 6

• Input date and store in variable / use directly
• Access all seven (indexes 0 to 6) events in array // loop for each event in array
• Attempt at selection…
• …to compare <code>date</code> input against <code>date</code> <u>in array</u> (element 0)
• …adding <code>length</code> (element 3) <u>from array</u> to the total <u>if dates match</u>.
• Outputting <u>calculated</u> total and date in appropriate message(s) <u>at the end</u></td><td>6
(AO3 2b)</td><td>BP2 can be achieved either by iteration accessing each event or manually repeating code to access each event. Must be 0 to 6, not 1 to 7.

Allow reference <code>to events</code> (table given) or <code>arrayEvents</code> (2D array) in answer as long as used consistently.

BP2 loop allow off by one errors (Python), looping to array length or array length – 1. Allow <code>for each item</code> in array or any other suitable loop.

BP4 and BP5 allow array reference as either column major or row major.

Output can either be once at the end or on every iteration, as long as it is output at the end.

Only give output mark if attempt made to calculate total <u>within the algorithm</u>.

Do not penalise capitalisation or minor misspellings of variable names.

<u>Example answer 1</u>

<pre>total = 0
date = input("Please enter
date")
for count = 0 to
events.length-1
  if events[0, count] == date
then
  total = total +
events[3,count]
  endif
next count
print("There were " + total +
" events on " + date)</pre>
<u>Example answer 2</u></td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance

					<pre>total = 0
					date = input("Please enter
					date")
					for item in events:
					  if item[0] == date then
					  total = total + item[3]
					  endif
					next count
					print("There were " + total +
					" events on " + date)</pre>
					Examiner's Comments
					
					The final question in Section B is expected to be a high demand question.
					
					The techniques required (iteration through a 2D array, selection, keeping a running total of times) are within the specification but it is acknowledged that the level of challenge was high.
					
					Examiners were instructed to give marks for an attempt at a solution (as with previous questions).
					
					For this question marks were given for:
					
					• any attempt at selection
					• any solution that accessed each element in the given array, even if this was via a manual process.
					
					Therefore, many candidates gained multiple marks for an attempt that only partially solved the problem.
					
					A significant number of candidates were able to create a solution that fully met the requirements of the question. This was often done in an elegant and efficient manner.

Question			Answer/Indicative content	Marks	Guidance

					This is extremely pleasing and shows excellent understanding and significant experience of practical programming.
					
					Exemplar 3
					
					total = 0
					date = input ("Enter a date")
					~~toto~~
					~~count = 0~~
					for count = 0 to arrayEvents.length
					    if arrayEvents [0, count] == date then
					        total = total + arrayEvents [3, count]
					    endif
					~~endfor~~
					
					print ("Sensors were activated for", total, "Seconds on", date)
					
					Exemplar 3 shows a high scoring response. A date has been asked for as input which has then been used to compare to each element at position 0 in the array.
					
					Where any of these match, the total variable is updated to keep a running total of the corresponding element at position 3 in the array.
					
					After each element has been checked, the total and date are output in a suitable message.
					
					This is not the only method by which a response could be given full marks but is perhaps the most common.

			Total	7	

Question			Answer/Indicative content	Marks	Guidance
4			• For loop changed to **include 0** • total = 0 moved to **before** loop starts / removed	2 (AO3 2c)	Allow loop changed to 0 to 8 or 0 to 9 (Python) Do not accept moving total outside loop, NE (could be moved to after loop which would still be a logic error). Do not accept move to top of loop. Accept corrected code shown. Accept reference to count variable limits for BP1. <u>**Examiner's Comments**</u> This question acted as an excellent discriminator. The majority of candidates identified and fixed the problem with the array (starting at 0 rather than the 1 given). Fewer candidates were able to identify and fix the issue relating to the total being set to zero for each iteration. A number of candidates identified this second issue but were either not precise enough with their response (suggesting that the line could be moved but not stating where it should be moved to) or did not attempt to go beyond the simple identification. Candidates should be clear that this question required detail of how the program could be refined to fix problems, not just to identify problem(s).
			Total	2	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>5</td><td></td><td></td><td>

Function call	Returned value
`checkblock(2,1)`	B
`checkblock(3,0)`	A
`checkblock(2,3)`	FREE

</td><td>3</td><td>Do not accept "blank" or any other returned value for third call.

Ignore case and spelling as long as recognisable.</td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>3</td><td></td></tr>
<tr><td>6</td><td></td><td></td><td><code>print (minsPlayed[0,4])</code></td><td>1
(AO3 2b)</td><td><u>High-level programming language / OCR Exam Reference Language response required</u>

Do not accept pseudocode / natural English.

<code>print</code> may be a suitable output command word that could be found in a HLL e.g. <code>print</code> (Python), <code>console.writeline</code> (VB), <code>cout</code> (C++)

The array elements may be accessed together <code>[0,4]</code> (VB.NET) or separately <code>[0][4]</code> (Python)</td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>1</td><td></td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
7	a	i	print (hoursPlayed[0,2])	1	Correct Answer Only
		ii		1	Correct Answer Only
		iii	80	1	Correct Answer Only
		iv	• Adding all correct elements • Outputting correctly • Using a loop e.g. total = 0 for x = 0 to 4 total = total + hoursPlayed[0,x] next x print (total)	3	1 mark per bullet to a maximum of 3. If used, a flowchart should represent the bulleted steps in the answer column

	b		• Loop 0 to 29
			• Loop 0 to 4
			• Accessing hoursplayed[x,y]
			• Addition of hoursplayed[x,y] to total
			• Calculating average correctly outside of loops
			• Outputting the results
			
			e.g.
			total = 0
			for x = 0 to 29
			   for y = 0 to 4
			      Total = total + hoursPlayed[x,y]
			  next y
			nextx
			average = total / (30*5)
			print (average)
				6	Accept any type of average calculation (mean, median, mode).
					
					If used, a flowchart should represent the bulleted steps in the answer column.

			Total	**14**	